1. Write a program to generate and plot the following sequences: (a) Unit sample sequence $\delta(n)$, (b) unit step sequence *u(n)*, (c) ramp sequence *r(n)*, (d) real valued exponential sequence $x(n) = (0.8)^n u(n)$ for $0 \le n \le 50$.
2. Write a program to compute the convolution sum of a rectangle signal (or gate function) with itself for N = 5

$$\mathrm{x(n)} = \mathrm{rect}\left(\frac{\mathrm{n}}{2\mathrm{N}}\right) = \prod \left(\frac{\mathrm{n}}{2\mathrm{N}}\right) = \{1 \quad -\mathrm{N} \le \mathrm{n} \le \mathrm{N} \; 0 \qquad \text{otherwise}$$

3. An LTI system is specified by the difference equation: y(n)=0.8y(n-1)+x(n)
   (a) Determine $\mathrm{H(e^{iw})}$
   (b) Calculate and plot the steady state response y(n) to $\mathrm{x(n)} = \cos\cos\,(0.5\pi\mathrm{n})\;\mathrm{u(n)}$
4. Given a casual system y(n)=0.9y(n-1)+x(n)
   (a) Find H(z) and sketch its pole-zero plot
   (b) Plot the frequency response $\left|H(e^{jw})\right|$ and $\angle H(e^{jw})$
5. Design a digital filter to eliminate the lower frequency sinusoid of *x(t)=sin7t+sin200t*. The sampling frequency is 500 Hz. Plot its pole zero diagram, magnitude response, input and output of the filter.
6. Let *x(n)* be a 4-point sequence:
   $x(n) = \{1,1,1,1\} = \{1 \;\; 0 \le n \le 3 \; 0 \;\; otherwise$

   Compute the DTFT $X(e^{jw})$ and plot its magnitude
   (a) Compute and plot the 4 point DFT of *x(n)*
   (b) Compute and plot the 8 point DFT of *x(n)* (by appending 4 zeros)
   (c) Compute and plot the 16 point DFT of *x(n)* (by appending 12 zeros)
7. Let *x(n)* and *h(n)* be the two 4-point sequences,
   x(n)={1,2,2,1} *h(n)={1,-1,-1,1}*

   Write a program to compute their linear convolution using circular convolution.
8. Using a rectangular window, design a FIR low-pass filter with a pass-band gain of unity, cut off frequency of 1000 Hz and working at a sampling frequency of 5 KHz. Take the length of the impulse response as 17.

9. Design an FIR filter to meet the following specifications:

   Passband edge F_p=2 KHz
   Stopband edge F_s=5 KHz
   Passband attenuation A_p=2 dB
   Stopband attenuation A_s=42 dB
   Sampling frequency F_{sf}=20 KHz

10. The frequency response of a linear phase digital differentiator is given by

$$\mathrm{H_d}\left(\mathrm{e^{jw}}\right) = \mathrm{jwe^{-j\tau w}} \;\; |\mathrm{w}| \le \pi$$

    Using a Hamming window of length M = 21, design a digital FIR differentiator. Plot the amplitude response